

Follow the instructions below:

1. The exam must be solved individually.
2. Submitting materials retrieved from the internet is plagiarism.
3. You may use all notebooks posted on the course web site and your own notebooks.
4. Solutions must be in a Jupyter notebook, with a SageMath kernel.
5. The questions are provided in a Jupyter notebook.
You may use that notebook to formulate your answers to the questions.
6. Answers must be submitted before, or at 1:50pm.
7. Submit to gradescope.
8. Not submitting any answers will by default result in a zero score.
9. During the exam no questions will be answered, so do not ask questions.

Good Luck!

1. Let $S = \sqrt{\pi/3}$.

- (a) Compute a sequence of nearby rational approximations for S where the size of the denominator of the k -th approximation is bounded by 10^k .
- (b) Compute the relative errors of those 20 rational approximations to verify the accuracy.

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2. Consider $p = x^4 + 3x^3 + 2x^2 + 2x + 4$ as a polynomial with coefficients in $\mathbb{Q}$.

Add sufficiently many formal roots to $\mathbb{Q}$ so that p can be written as the product of four linear factors. What are the roots in the extended field?

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3. What is delayed function evaluation? How to execute the evaluation of such a function call? Illustrate the difference with a good example from the course.

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4. Consider the function $f(n) = \frac{2}{n} \sum_{k=1}^{n-1} \left(\frac{k}{n}\right)^2 \cos\left(\frac{k}{n}\right)$.

- (a) Code the definition of f into a list comprehension to make a function F which for some positive integer n returns the floating-point approximation of $f(n)$ in double precision.
- (b) Time the execution of F for $n = 10,000$. Explain why F is inefficient.
- (c) Use numpy to vectorize F , as defined in the function $V(n)$. Verify that V is correct. For $n = 10,000$, compare the running time of V with F .

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5. Make a fast callable object of $x - \frac{y^2}{2x + y^2 - 3}$ and draw its expression tree.

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6. Let $q = \frac{4.12x^6 - 1}{x^6 - 1}$ be a polynomial with complex coefficients.

- (a) Compute the partial fraction decomposition p of q . Relate the denominators of the partial fraction decomposition p to the roots of the denominator of q .
- (b) Given the partial fraction decomposition p of q , reconstruct q . Describe what you observe when you normalize.

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7. Transform $p = (2y - z)^6 - x^4$ into $r = ((2y - z)^3 + x^2)((2y - z)^3 - x^2)$, without typing r .

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